



Ex.No.7 - Motion Tracking in Video

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In [5]: import cv2
import numpy as np
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In [9]: # Direct video file path
video_path = r"D:\VI\IP & CV\Lab Experiments\Inputs\Traffic Road.mp4"

# Open video file
cap = cv2.VideoCapture(video_path)

if not cap.isOpened():
    print(f"Error: Could not open video file '{video_path}'")
    print("\nTroubleshooting tips:")
    print("1. Make sure the file exists at that location")
    print("2. Try using forward slashes instead: C:/Users/MADHU/Downloads/t1.m")
    print("3. Check if the file is corrupted")
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In [10]: # Read first frame to initialize background
ret, first_frame = cap.read()

if not ret:
    print("Error: Could not read the video")
else:
    prev_gray = cv2.cvtColor(first_frame, cv2.COLOR_BGR2GRAY)
    prev_gray = cv2.GaussianBlur(prev_gray, (15, 15), 0)

print("Motion Tracking Started...")
print("Press 'q' to quit\n")
```

Motion Tracking Started...
Press 'q' to quit

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In [11]: while True:

    ret, frame = cap.read()
    if not ret:
        break

    # Convert frame to grayscale
    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Blur frame
    gray = cv2.GaussianBlur(gray, (15, 15), 0)
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# Difference between frames
frame_diff = cv2.absdiff(prev_gray, gray)

# Threshold
thresh = cv2.threshold(frame_diff, 15, 255, cv2.THRESH_BINARY)[1]

# Dilate image
thresh = cv2.dilate(thresh, None, iterations=3)

# Find contours
contours, _ = cv2.findContours(
    thresh.copy(),
    cv2.RETR_EXTERNAL,
    cv2.CHAIN_APPROX_SIMPLE
)

motion_detected = False

for contour in contours:

    area = cv2.contourArea(contour)

    if area > 200:
        motion_detected = True

        x, y, w, h = cv2.boundingRect(contour)

        cv2.rectangle(frame, (x, y), (x+w, y+h), (0,255,0), 3)

status = "MOTION DETECTED" if motion_detected else "No Motion"
color = (0,255,0) if motion_detected else (0,0,255)

cv2.putText(frame, status, (10,30),
            cv2.FONT_HERSHEY_SIMPLEX, 1, color, 2)

cv2.putText(frame, f"Contours: {len(contours)}", (10,70),
            cv2.FONT_HERSHEY_SIMPLEX, 0.7, (255,255,255), 2)

cv2.imshow("Motion Tracking", frame)
cv2.imshow("Frame Difference", frame_diff)
cv2.imshow("Threshold", thresh)

prev_gray = gray

if cv2.waitKey(30) & 0xFF == ord('q'):
    break

cap.release()
cv2.destroyAllWindows()
print("Motion Tracking Stopped")

```

Motion Tracking Stopped

